Few Pixels, Fragile Faithfulness: Evaluating Visual Explanations under Sparse Perturbations

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Abstract. Visual explanations can be manipulated by input perturbations, raising concerns about their trustworthiness. Existing explanation-shift attacks primarily focus on changing the saliency map while preserving the model's prediction, but such shifts do not necessarily degrade explanation faithfulness. In this work, we propose a sparse, prediction-preserving attack that jointly optimizes explanation shift and faithfulness reduction. To handle the discrete nature of sparse perturbations, we employ a gradient-free attack based on a genetic algorithm, using surrogate objectives for insertion, deletion, and saliency shift. Extensive experiments with multiple popular vision models and widely-used visual explanation methods show that, by modifying only 0.1% of the image pixels, our attack can redirect saliency toward less relevant regions and substantially degrade explanation faithfulness while preserving the original prediction. Our results reveal a key limitation of existing explanation-shift attacks and highlight an important concern regarding the trustworthiness of visual explanations: even under a stringent sparsity constraint, explanations can be substantially shifted while their faithfulness is simultaneously compromised. The source code is available at the following link: https://github.com/khoa16122004/Shift-Faithfulness-Attack.

Keywords: Visual Explanation · Adversarial Robustness · Explanation Faithfulness

1 Introduction

As deep neural networks (DNNs) become increasingly influential in visual recognition, understanding the reasoning behind their predictions is essential for assessing the trustworthiness of vision models. Accordingly, visual explanation methods are used to provide insights into a model's predictions by producing a saliency map that highlights image regions considered important for its decision [1, 17, 18, 20, 21, 26, 29]. Beyond explaining the model predictions, it is equally important to assess whether these explanations can be trusted, particularly in safety-critical domains such as medical imaging. A common way to probe this trustworthiness is through adversarial attacks that preserve the model's prediction while shifting the saliency map to different regions [4, 9]. We refer to this type of attack as an *explanation-shift attack*. However, these conventional explanation-shift attacks [4, 9], rely almost entirely on gradient-based, dense perturbations, where modifications are allowed to be distributed across the entire image. In

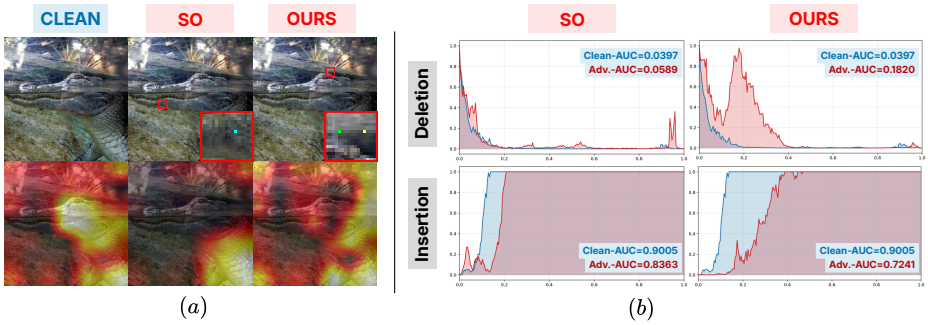


Fig. 1. Comparison of Different Explanation Attacks. **(a)** Visualization of the clean input and saliency map (**Clean**), the adversarial input and saliency map generated by the Shift-Only (**SO**) attack, which optimizes only the explanation-shift objective, and those generated by our proposed method (**Ours**), which jointly optimizes the faithfulness-reduction and explanation-shift objectives. The highlighted regions indicate sparsely perturbed locations, with one region magnified for closer inspection. **(b)** Faithfulness evaluation of the clean and adversarial saliency maps using the corresponding Deletion (top) and Insertion (bottom) curves. For each attack, the red curve corresponds to the adversarial image, while the blue curve corresponds to the corresponding clean input.

contrast, *sparse perturbations*, which constrain the number of modified pixels, remain less explored, partly because sparse pixel selection introduces a discrete optimization challenge [2], together with the additional complexity of optimizing through both the model and the explanation function [4, 9]. A recent sparse attack has been proposed [24] to investigate the stability of saliency maps under sparse perturbations. However, the faithfulness of the resulting explanations is not explicitly considered [4, 9, 24], leaving open the question of whether the induced saliency shift actually compromises explanation faithfulness.

To examine this issue, we evaluate the faithfulness of saliency maps under a sparse adversarial attack [24]. Our results reveal a key limitation of existing explanation-shift attacks: **shifting the saliency map does not necessarily make the resulting explanation less faithful**. Faithfulness measures whether the features identified by an explanation method accurately reflect the evidence underlying the model's prediction. It is commonly evaluated using Deletion and Insertion, where a faithful explanation should cause the prediction confidence to decrease rapidly when salient regions are removed and increase rapidly when they are restored [17]. As shown in Fig. 1(a), an attack optimized purely for explanation shift, denoted as the Shift-Only (SO) attack, can relocate the salient regions, yet the newly highlighted regions may still reflect what the model uses to make its decision, so the Deletion/Insertion faithfulness curves barely change (see Fig. 1(b), left). In other words, shifting where an explanation points does not necessarily compromise its faithfulness to the model's decision. Therefore, a stronger attack should instead redirect saliency toward regions that are less relevant to the prediction, thereby degrading faithfulness while simultaneously inducing an explanation shift, which is precisely what our method achieves in Fig. 1(a–b).

Given these limitations, we evaluate explanation methods under sparse perturbations by considering not only saliency shift, as in prior explanation-shift studies, but also the faithfulness of the resulting explanations. We propose a sparse explanation attack that jointly optimizes **explanation shift** and **faithfulness reduction**, using a gradient-free attack based on a genetic algorithm to handle the discrete nature of sparse perturbations. Across six vision models and six visual explanation methods, our proposed attack, by modifying only 0.1% of the total pixels, consistently outperforms the Shift-Only baseline in reducing explanation faithfulness while inducing a substantial saliency shift without changing the model's prediction, as illustrated in Fig. 1. These results highlight an important concern regarding the trustworthiness of visual explanations by showing that sparse perturbations can simultaneously shift explanations and compromise their faithfulness.

2 Related Work

Visual Explanation for Vision Models. Early studies on visual explanation investigated how individual input pixels or image regions contribute to a model's prediction by generating a heat map, commonly referred to as a *saliency map* [6,8,17,20]. These methods provide a visual representation of the model's decision by assigning importance scores to different image regions, and have been widely studied for convolutional neural networks (CNNs). With the introduction of self-attention in Vision Transformers (ViTs) [5], subsequent studies extended visual explanation methods to Transformer-based architectures by analyzing attention weights and their propagation across layers [1, 18].

Adversarial Attacks on Explanation. Existing studies have revealed limitations of saliency maps, including failures under sanity checks and inconsistencies across attribution methods [11, 14, 27]. Moreover, visual explanations may be susceptible to small input changes even when the prediction remains unchanged. For example, gradient-based attacks such as PGD [16] have been shown to substantially suppress or relocate salient regions without changing the predicted class [9]. Other studies further demonstrate that explanations can be manipulated toward predefined target patterns through gradient-based optimization [4]. More recently, sparse explanation attacks have explored discrete perturbations by modifying only a limited number of pixels [24]. Existing explanation-shift attacks focus primarily on shifting saliency maps and do not explicitly evaluate the faithfulness of the resulting explanations, leaving it unclear whether the induced saliency shifts actually compromise explanation faithfulness.

3 Methodology

This section presents our proposed sparse attack for simultaneously inducing explanation shift and reducing explanation faithfulness. An overview of the attack pipeline is illustrated in Figure 2.

3.1 Formulation

Given an image $x \in [0,1]^{H\times W\times 3}$, let $Y = \{1, 2, \ldots, L\}$ denote the set of classes. We consider a classification model $f : x \mapsto [0,1]^L$, which outputs a probability

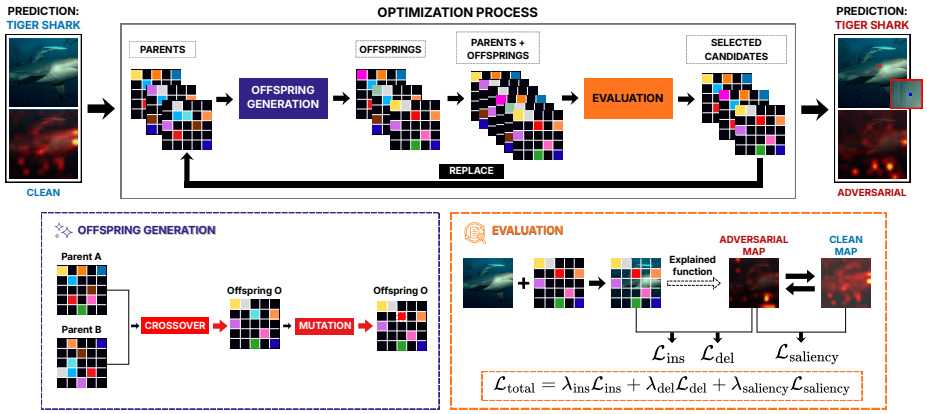


Fig. 2. Overview of the sparse attack algorithm. Starting from an initial population of sparse perturbation candidates, the optimization process iteratively evolves the candidates to identify a solution that shifts the saliency map and reduces explanation faithfulness.

distribution over the L classes. We assume that the model outputs softmax-normalized probabilities. The predicted class is defined as $\hat{y} = \arg\max_{l \in Y} f(x)_l$, where $f(x)_l$ denotes the predicted probability of class l. To explain the model's prediction, we employ a visual explanation method G (e.g., Grad-CAM [20], Rollout [1]). Given an input image x, the classification model f, and a target class $\hat{y}$, the explainer produces a saliency map $\phi^x_{\hat{y}} = G(x, f, \hat{y}) \in [0,1]^{H \times W}$, where each value indicates the relative importance of the corresponding image region to the model's output for class $\hat{y}$. A higher saliency value indicates that the corresponding region is assigned greater importance by the explanation method.

To evaluate the faithfulness of the saliency map $\phi^x_{\hat{y}}$ generated by the explanation method G, we consider the Deletion and Insertion metrics [17]. Both metrics iteratively manipulate the input according to the saliency ranking induced by $\phi^x_{\hat{y}}$ and measure the resulting prediction confidence $f(\cdot)_{\hat{y}}$. Specifically, Deletion progressively removes the most salient regions from x, while Insertion progressively restores the most salient regions from x into a baseline image b. Let $x_{\text{del}}(p)$ and $x_{\text{ins}}(p)$ denote the corresponding images after processing a fraction $p \in [0,1]$ of the pixels. The two metrics are defined as:

$$\text{DeletionAUC}(x, \phi^x_{\hat{y}}, f) = \int_0^1 f(x_{\text{del}}(p))_{\hat{y}}\, dp, \tag{1}$$

$$\text{InsertionAUC}(x, \phi^x_{\hat{y}}, f) = \int_0^1 f(x_{\text{ins}}(p))_{\hat{y}}\, dp. \tag{2}$$

A faithful saliency map is expected to produce a lower DeletionAUC and a higher InsertionAUC, indicating that the highlighted regions have a stronger influence on the model's prediction.

Our objective is to find a sparse perturbation $\delta \in [-1,1]^{H \times W \times 3}$ that shifts saliency from important regions toward less important regions, thereby reducing the faithfulness

of the resulting explanation. The perturbed image $x + \delta$ may contain values outside the valid image range $[0, 1]$; therefore, we define the adversarial image as $x_{\text{adv}} = \text{clip}(x + \delta, 0, 1)$, where the clipping operation ensures that all pixel values of x_{adv} remain within $[0, 1]$. Let $\phi_{\hat{y}}^{x_{\text{adv}}} = G(x_{\text{adv}}, f, \hat{y})$ denote the saliency map generated for the adversarial image with respect to the original predicted class $\hat{y}$. To explicitly measure the sparsity of the perturbation at the pixel level, we define the set of perturbed spatial locations as $C(\delta) = \{(i, j) | \delta_{i,j} \neq \mathbf{0}\}$, where $\delta_{i,j}$ denotes the perturbation applied to the pixel at spatial location (i, j). Our problem can be formulated as:

$$\begin{aligned} \arg\min_{\delta} \quad & \underbrace{\text{InsertionAUC}(x_{\text{adv}}, \phi_{\hat{y}}^{x_{\text{adv}}}, f) - \text{DeletionAUC}(x_{\text{adv}}, \phi_{\hat{y}}^{x_{\text{adv}}}, f)}_{\text{Faithfulness Reduction}} \\ & + \underbrace{\text{SpearmanCorr}\left(\phi_{\hat{y}}^{x_{\text{adv}}}, \phi_{\hat{y}}^{x}\right)}_{\text{Explanation Shift}}, \\ \text{s.t.} \quad & |C(\delta)| \leq \epsilon, \qquad \arg\max_{l} f(x + \delta)_l = \hat{y}, \end{aligned} \tag{3}$$

where the constraint $|C(\delta)| \leq \epsilon$ limits the maximum number of pixel locations that can be modified during the attack, thereby enforcing the sparsity of the perturbation. The prediction-preserving constraint $\arg\max_l f(x + \delta)_l = \hat{y}$ ensures that the adversarial image maintains the original prediction $\hat{y}$ after perturbation. InsertionAUC and DeletionAUC [17] measure the faithfulness of the saliency map through the insertion and deletion processes, respectively, while SpearmanCorr [23] measures the rank correlation between the adversarial and original saliency maps, thereby capturing the degree of explanation shift.

For ease of notation, we denote $\phi_{\hat{y}}^{x}$ and $\phi_{\hat{y}}^{x_{\text{adv}}}$ as ϕ^{x} and $\phi^{x_{\text{adv}}}$, respectively, throughout the following section. Unless otherwise specified, ϕ^{x} and $\phi^{x_{\text{adv}}}$ denote the saliency maps corresponding to the original predicted class $\hat{y}$ for the clean and adversarial images, respectively. $\phi_{i,j}^{x}$ and $\phi_{i,j}^{x_{\text{adv}}}$ denote the saliency values at spatial location (i, j) of the corresponding saliency maps.

3.2 Attack Objective

Although Insertion AUC, Deletion AUC, and Spearman correlation directly quantify the desired attack outcomes, these metrics are not ideal optimization objectives. Insertion and Deletion AUCs evaluate the model response through a progressive process based on the ranking and selection of salient regions, while Spearman correlation measures the agreement between the original and adversarial saliency maps based on their relative ordering. Consequently, these metrics rely primarily on ranking-based signals and provide limited information about how individual regions should be modified to induce the desired changes, resulting in relatively coarse guidance for optimizing sparse perturbations.

Therefore, to solve the problem defined in Equation 3, rather than directly optimizing these evaluation metrics, we construct surrogate objectives that provide smoother and

more informative signals for optimization. We define the attack objective as a combination of two complementary objectives: faithfulness reduction and explanation shift. Specifically, following [28], we use $\mathcal{L}_{\text{ins}}$ and $\mathcal{L}_{\text{del}}$, which provide smooth approximations of the insertion and deletion behaviors, respectively. To induce an explanation shift, we further employ a Jaccard-based objective [7], $\mathcal{L}_{\text{saliency}}$, which measures the overlap between the original and adversarial saliency maps.

Faithfulness Reduction Objective. Following [28], we leverage a faithfulness-reduction loss that provides a smooth approximation of the conventional InsertionAUC and DeletionAUC.

Given the saliency map $\phi^{x_{\text{adv}}}$, we first flatten and sort its values in descending order, yielding $\tilde{\phi}^{x_{\text{adv}}}$. For each step s of the insertion or deletion process, from $\tilde{\phi}^{x_{\text{adv}}}$, we define the threshold $t_s = \frac{\tilde{\phi}_s^{x_{\text{adv}}} + \tilde{\phi}_{s+1}^{x_{\text{adv}}}}{2}$, which lies midway between the last selected saliency value and the first unselected saliency value. Then the soft mask for each step can be defined as:

$$r(\phi_{ij}^{x_{\text{adv}}}; s) = \sigma\left(T(\phi_{ij}^{x_{\text{adv}}} - t_s)\right), \tag{4}$$

where $\sigma(\cdot)$ denotes the sigmoid function and T controls the sharpness of the transition. Thus, saliency values substantially larger than t_s produce mask values close to 1, while those substantially smaller than t_s produce values close to 0.

To reduce computational cost, rather than evaluating the insertion and deletion processes at every individual pixel, we divide the $N = HW$ spatial locations into S steps. Each step therefore incorporates approximately N/S pixels. Specifically, letting $q = \max\left(1, \left\lfloor \frac{N}{S} \right\rfloor\right)$, the threshold at step s is computed as

$$t_s = \frac{\tilde{\phi}_{sq-1}^{x_{\text{adv}}} + \tilde{\phi}_{sq}^{x_{\text{adv}}}}{2}, \qquad s = 1, \ldots, S. \tag{5}$$

Let $b \in \mathbb{R}^{H \times W \times 3}$ denote a black baseline image, represented by a zero tensor. We then define the operators for generating the masked image at each step of the faithfulness evaluation as

$$\begin{aligned} \alpha_{\text{soft}}(\phi^{x_{\text{adv}}}, b, s) &= r(\phi^{x_{\text{adv}}}, s) \odot x_{\text{adv}} + (1 - r(\phi^{x_{\text{adv}}}, s)) \odot b, \\ \beta_{\text{soft}}(\phi^{x_{\text{adv}}}, b, s) &= r(\phi^{x_{\text{adv}}}, s) \odot b + (1 - r(\phi^{x_{\text{adv}}}, s)) \odot x_{\text{adv}}, \end{aligned} \tag{6}$$

From this formulation, we define the following two minimization losses for the insertion and deletion processes:

$$\mathcal{L}_{\text{ins}} = \frac{1}{S} \sum_{s=1}^{S} f\left(\alpha_{\text{soft}}(\phi^{x_{\text{adv}}}, b, s)\right)_{\hat{y}}, \quad \mathcal{L}_{\text{del}} = -\frac{1}{S} \sum_{s=1}^{S} f\left(\beta_{\text{soft}}(\phi^{x_{\text{adv}}}, b, s)\right)_{\hat{y}}. \tag{7}$$

Explanation Shift Objective. To measure the difference between the original and adversarial saliency maps, we compute their Jaccard similarity [7]:

$$\mathcal{L}_{\text{saliency}} = \frac{\sum_{i=1}^{H}\sum_{j=1}^{W} \min\left(\phi_{i,j}^{x}, \phi_{i,j}^{x_{\text{adv}}}\right)}{\sum_{i=1}^{H}\sum_{j=1}^{W} \max\left(\phi_{i,j}^{x}, \phi_{i,j}^{x_{\text{adv}}}\right)}. \tag{8}$$

Overall Objective. We combine the faithfulness-reduction and explanation-shift objectives into a unified optimization objective:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{ins}}\mathcal{L}_{\text{ins}} + \lambda_{\text{del}}\mathcal{L}_{\text{del}} + \lambda_{\text{saliency}}\mathcal{L}_{\text{saliency}}, \tag{9}$$

where $\lambda_{\text{ins}}, \lambda_{\text{del}}, \lambda_{\text{saliency}} \in \{0, 1\}$. The **Shift-Faithfulness** attack variant jointly optimizes all three objectives with $(\lambda_{\text{ins}}, \lambda_{\text{del}}, \lambda_{\text{saliency}}) = (1, 1, 1)$ and serves as our proposed attack. We consider the following alternative variants by activating different objective combinations: **Deletion-Driven** $(0, 1, 0)$, **Insertion-Driven** $(1, 0, 0)$, **Faithfulness-Driven** $(1, 1, 0)$, and **Shift-Only** $(0, 0, 1)$.

3.3 Sparse Attack Algorithm

Due to the discrete nature of sparse perturbations [2], as well as the complexity of optimizing through the saliency map and explanation function, we adopt a gradient-free sparse attack based on a genetic algorithm, inspired by [24]. The overall attack procedure is illustrated in Fig. 2.

Initialization. Following Sparse-RS [3,25], we initialize a population of P candidate perturbations, each containing $k = \epsilon$ perturbation locations. The locations are randomly sampled from all image locations, while the perturbation value at each selected location is independently sampled for each color channel from $\{-1, 0, 1\}$. The probability of assigning zero perturbation is controlled by p_0, with the remaining probability equally distributed between -1 and 1.

Population Evolution. At each iteration, candidate solutions from the current population are selected as parents to generate offspring through crossover and mutation operators. These operators modify the perturbation locations and their corresponding values while preserving the perturbation budget. The newly generated candidates are then evaluated using $\mathcal{L}_{\text{total}}$ and compared with the existing population. A selection procedure is then applied to retain the candidates with better objective values, forming the population for the next iteration. This process is repeated for a predefined number of iterations. Throughout the optimization, we maintain the candidate with the lowest $\mathcal{L}_{\text{total}}$ among those satisfying all attack constraints.

Final Selection. At the end of the optimization process, we select the candidate with the lowest $\mathcal{L}_{\text{total}}$ among those satisfying all attack constraints as the final adversarial perturbation.

4 Experiment

4.1 Experiment Setup

Target Vision Models and Explanation Methods. We conduct our experiments on six vision models, including ResNet-18 [10], VGG-16 [22], and DenseNet-121 [12] as CNN-based models, and ViT-B/32, ViT-B/16 [5], and SwinV2-Tiny [15] as transformer-based models. For visual explanation methods, we employ Grad-CAM [20], HiResCAM [6], LayerCAM [13], and XGrad-CAM [8] for CNN-based models, while Raw Attention and Attention Rollout [1] are applied to the transformer-based models.

Datasets. We evaluate our method on the ImageNet-1K dataset [19]. Specifically, for each vision model, we randomly select 100 samples that are correctly classified by the model. These samples are then used to evaluate the effectiveness of our attack method and compare it against the baseline variants.

Baselines. Shift-Faithfulness is our proposed attack, which jointly optimizes all three objectives with $(\lambda_{\text{ins}}, \lambda_{\text{del}}, \lambda_{\text{saliency}}) = (1, 1, 1)$. We consider the following variants, obtained by activating different combinations of the objectives: **Deletion-Driven** $(0, 1, 0)$, **Insertion-Driven** $(1, 0, 0)$, **Faithfulness-Driven** $(1, 1, 0)$, and **Shift-Only** $(0, 0, 1)$.

Attack Hyperparameters. Following previous work [3, 24], we set the sparse perturbation budget to $\epsilon = 50$, corresponding to approximately 0.1% of the pixels in a 224×224 image. For the sparse attack algorithm, we use a population size of 100 and perform 200 optimization iterations, with the probability of assigning a zero perturbation to a pixel channel set to $p_0 = 0.3$. For the faithfulness reduction objective, we set the number of progressive steps S in $\mathcal{L}_{\text{ins}}$ and $\mathcal{L}_{\text{del}}$ to 224.

Evaluation Metrics. We report the Area Under the Curve (AUC) of the Deletion and Insertion curves [17], denoted as DeletionAUC and InsertionAUC, respectively. We further compute the Insertion Minus Deletion (IMD) score as IMD = InsertionAUC − DeletionAUC, where a lower IMD indicates weaker faithfulness. To measure explanation shift, we use the Spearman Rank Correlation [23], denoted as SpearmanCorr, between the original and adversarial saliency maps, where a lower value indicates a larger shift.

4.2 Main results

Table 1 presents the faithfulness evaluation results for CNN and transformer-based models. Our results show that sparse, prediction-preserving perturbations can substantially shift saliency toward less relevant regions while simultaneously reducing the faithfulness of the resulting explanations. Specifically, compared with the clean saliency maps (Clean), our proposed Shift-Faithfulness attack achieves the lowest IMD across all evaluated cases, as indicated by the red cells. For example, for Grad-CAM on VGG-16, IMD decreases from 0.6107 (Clean) to 0.1619 (Shift-Faithfulness), while for XGrad-CAM on DenseNet-121, it decreases from 0.3961 to −0.0105.

Compared with the Shift-Only baseline, which can affect the faithfulness of the resulting explanations to some extent, the Shift-Faithfulness attack achieves substantially larger reductions in IMD across the evaluated settings. For instance, on DenseNet-121 with HiResCAM, Shift-Only only slightly reduces IMD from 0.5709 under the

Table 1. Faithfulness evaluation under different optimization objectives on CNN and ViT models. We report DeletionAUC (Del.)↑, InsertionAUC (Ins.)↓, and IMD↓, where the indicated direction corresponds to weaker faithfulness.

CNN Models

		ResNet18			VGG16			DenseNet121		
Method	**Setting**	Del.	Ins.	IMD	Del.	Ins.	IMD	Del.	Ins.	IMD
GradCAM	Clean	0.1496	0.8030	0.6533	0.1761	0.7867	0.6107	0.1764	0.7455	0.5691
	Shift-Only	0.1332	0.6626	0.5294	0.1923	0.6081	0.4157	0.1440	0.6824	0.5383
	Deletion-Driven	0.2245	0.7727	0.5482	0.4015	0.7056	0.3040	0.2648	0.7281	0.4633
	Insertion-Driven	0.1554	0.5804	0.4249	0.2177	0.4895	0.2717	0.1423	0.5662	0.4239
	Faithfulness-Driven	0.2081	0.5770	0.3689	0.3531	0.5843	0.2312	0.2035	0.6033	0.3997
	Shift-Faithfulness	0.1964	0.5485	0.3521	0.3803	0.5422	0.1619	0.2182	0.5305	0.3124
HiresCAM	Clean	0.1496	0.8030	0.6534	0.1737	0.7815	0.6078	0.1768	0.7477	0.5709
	Shift-Only	0.1332	0.6625	0.5293	0.1737	0.6590	0.4854	0.1300	0.6973	0.5673
	Deletion-Driven	0.2250	0.7628	0.5378	0.3484	0.7293	0.3809	0.2634	0.7230	0.4596
	Insertion-Driven	0.1544	0.5844	0.4300	0.1930	0.5268	0.3338	0.1452	0.6001	0.4550
	Faithfulness-Driven	0.2021	0.5631	0.3610	0.3121	0.6328	0.3207	0.2295	0.5682	0.3387
	Shift-Faithfulness	0.1985	0.5393	0.3409	0.3086	0.6025	0.2939	0.2300	0.5567	0.3267
LayerCAM	Clean	0.1515	0.7937	0.6421	0.1454	0.8106	0.6652	0.1801	0.7341	0.5540
	Shift-Only	0.1294	0.7298	0.6004	0.1302	0.7417	0.6115	0.1439	0.7025	0.5587
	Deletion-Driven	0.2323	0.7656	0.5333	0.2469	0.7796	0.5327	0.2634	0.7230	0.4596
	Insertion-Driven	0.1536	0.6007	0.4471	0.1259	0.6883	0.5624	0.1366	0.5740	0.4374
	Faithfulness-Driven	0.2041	0.6069	0.4027	0.1949	0.6638	0.4689	0.2295	0.5682	0.3387
	Shift-Faithfulness	0.1967	0.5698	0.3731	0.2137	0.6506	0.4369	0.2005	0.5711	0.3706
XGradCAM	Clean	0.1496	0.8030	0.6534	0.1684	0.7988	0.6304	0.2796	0.6756	0.3961
	Shift-Only	0.1332	0.6626	0.5294	0.1771	0.6489	0.4718	0.3676	0.5868	0.2192
	Deletion-Driven	0.2210	0.7709	0.5499	0.3742	0.7410	0.3668	0.4858	0.5162	0.0305
	Insertion-Driven	0.1540	0.5818	0.4279	0.2099	0.5498	0.3399	0.4051	0.4597	0.0546
	Faithfulness-Driven	0.2151	0.5717	0.3566	0.3206	0.6441	0.3235	0.4523	0.4915	0.0392
	Shift-Faithfulness	0.2161	0.5372	0.3211	0.3179	0.5863	0.2685	0.4826	0.4721	-0.0105

ViT Models

		ViT-B/32			ViT-B/16			Swin V2-Tiny		
Method	**Setting**	Del.	Ins.	IMD	Del.	Ins.	IMD	Del.	Ins.	IMD
Raw Attention	Clean	0.2810	0.5120	0.2310	0.3636	0.5538	0.1902	0.4312	0.6629	0.2317
	Shift-Only	0.3757	0.4371	0.0615	0.3875	0.4688	0.0812	0.5653	0.6162	0.0509
	Deletion-Driven	0.3884	0.4765	0.0881	0.4648	0.5154	0.0506	0.6132	0.6332	0.0200
	Insertion-Driven	0.3303	0.4318	0.1015	0.3730	0.4584	0.0855	0.4650	0.6183	0.1534
	Faithfulness-Driven	0.3535	0.3916	0.0381	0.4373	0.4659	0.0286	0.5872	0.6088	0.0216
	Shift-Faithfulness	0.3918	0.4193	0.0275	0.4540	0.4286	-0.0253	0.6083	0.5891	-0.0191
Rollout	Clean	0.2115	0.6234	0.4118	0.2192	0.6722	0.4530	0.3435	0.6897	0.3462
	Shift-Only	0.2726	0.5646	0.2920	0.2447	0.5715	0.3267	0.4458	0.7234	0.2777
	Deletion-Driven	0.2877	0.6021	0.3144	0.2982	0.6560	0.3578	0.5030	0.6919	0.1889
	Insertion-Driven	0.2307	0.5505	0.3198	0.2386	0.5804	0.3418	0.3525	0.6650	0.3126
	Faithfulness-Driven	0.2641	0.5492	0.2851	0.2848	0.5722	0.2874	0.4552	0.6745	0.2193
	Shift-Faithfulness	0.2875	0.5435	0.2560	0.2965	0.5562	0.2597	0.4814	0.6574	0.1760

Clean condition to 0.5673, whereas Shift-Faithfulness reduces it to 0.3267. A similar trend is observed with LayerCAM, where Shift-Only even increases IMD from 0.5540 to 0.5587, while Shift-Faithfulness substantially decreases it to 0.3706. These results demonstrate that **shifting the saliency map alone is insufficient to consistently undermine explanation faithfulness**, whereas jointly optimizing saliency shift and faithfulness reduction enables our attack to substantially reduce the faithfulness of the resulting explanations. Moreover, Shift-Faithfulness still induces a substantial explanation shift, although smaller than that achieved by Shift-Only, as shown in Fig. 3.

We provide additional qualitative results comparing our Shift-Faithfulness attack with the Shift-Only (SO) baseline in Fig. 4. Consider the Gila-monster sample shown in the top row. The SO attack shifts saliency away from the original salient region around the lizard's head. However, the shifted explanation still focuses on the lizard and

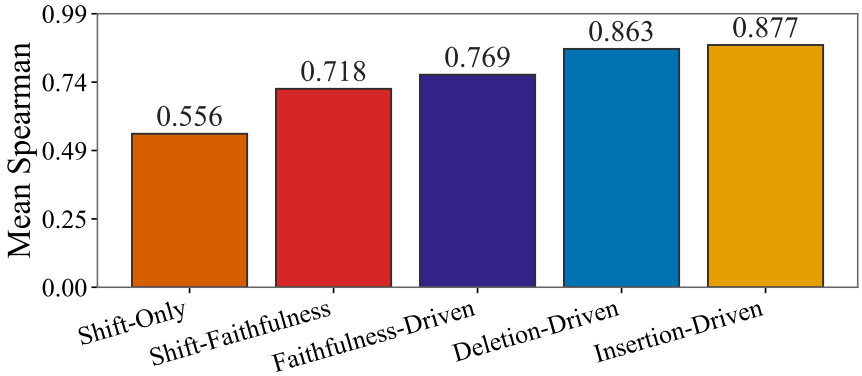


Fig. 3. Spearman rank correlation between the original and adversarial saliency maps across different attack variants. Lower values indicate a larger explanation shift.

does not reduce faithfulness in this example. In fact, its DeletionAUC decreases from 0.2622 to 0.0607, while its InsertionAUC increases from 0.8269 to 0.8987, indicating an improvement rather than a degradation in faithfulness. In contrast, our method redirects saliency from the lizard's head toward a less relevant region around its body and the surrounding background, resulting in a substantial explanation shift, with DeletionAUC increasing to 0.6933 and InsertionAUC decreasing to 0.4707.

4.3 Comparison of Different Attack Variants

In Figure 5, we present the trajectories of the three objective scores during optimization. First, when comparing the Deletion-Driven and Insertion-Driven variants, we observe a mild conflict between the two objectives. Specifically, optimizing $\mathcal{L}_{\text{del}}$ alone substantially reduces $\mathcal{L}_{\text{del}}$, while $\mathcal{L}_{\text{ins}}$ changes only marginally throughout optimization. Conversely, optimizing $\mathcal{L}_{\text{ins}}$ effectively reduces $\mathcal{L}_{\text{ins}}$, but $\mathcal{L}_{\text{del}}$ tends to increase. Furthermore, jointly optimizing these two objectives is more effective, enabling both $\mathcal{L}_{\text{del}}$ and $\mathcal{L}_{\text{ins}}$ to decrease simultaneously. This indicates that the two objectives are not naturally aligned: optimizing either independently provides limited improvement to the other and may even compromise it. For $\mathcal{L}_{\text{saliency}}$, optimizing the deletion or insertion objective independently still induces some saliency shift; however, the shift remains limited compared with the Faithfulness-Driven variant, which jointly optimizes both objectives.

By incorporating the saliency-shift objective, Shift-Faithfulness provides a better balance among the three objectives. Specifically, it effectively reduces both $\mathcal{L}_{\text{del}}$ and $\mathcal{L}_{\text{ins}}$ and achieves the second-lowest $\mathcal{L}_{\text{saliency}}$, following Shift-Only. This balanced optimization leads to the lowest IMD scores in Table 1, substantially outperforming the Faithfulness-Driven variant, which optimizes only the two faithfulness objectives. Further evidence is provided by the Deletion and Insertion curves in Fig. 6. Shift-Faithfulness

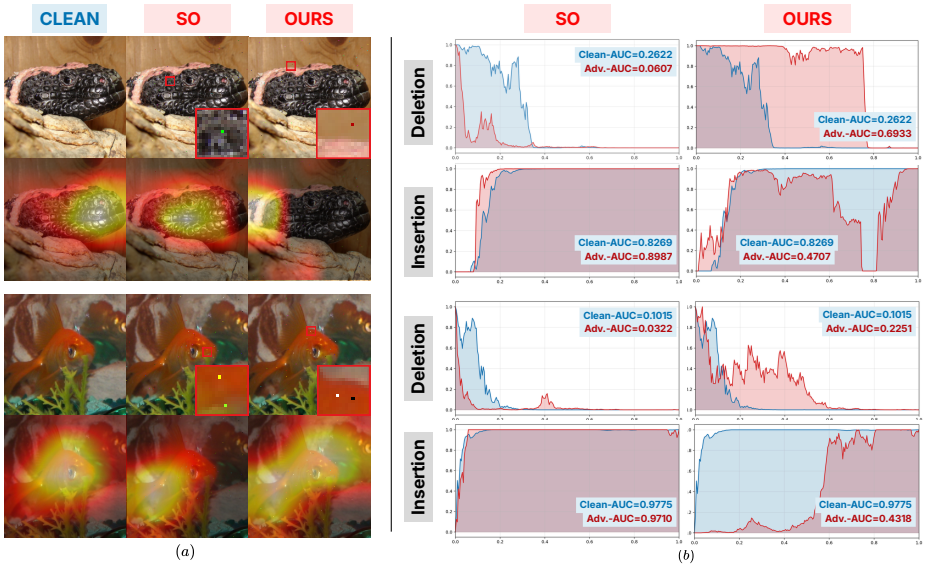


Fig. 4. Qualitative comparison of different attacks. **(a)** Visualization of the clean input and saliency map (**Clean**), the saliency map produced by the Shift-Only (**SO**) attack, and the saliency map produced by our proposed Shift-Faithfulness (**Ours**) attack. The highlighted regions indicate sparsely perturbed locations, with one region magnified for closer inspection. **(b)** Corresponding Deletion (top) and Insertion (bottom) curves for the clean and adversarial saliency maps, illustrating the changes in explanation faithfulness induced by the attacks.

consistently achieves higher Deletion curves than Deletion-Driven, while attaining one of the lowest Insertion curves among all attack variants. These results demonstrate that **integrating the explanation-shift objective with faithfulness reduction enables the simultaneous degradation of explanation faithfulness** while inducing a substantial saliency shift.

5 Conclusion

In this work, we investigated the vulnerability of post-hoc visual explanations to sparse, prediction-preserving perturbations and showed that explanation shift alone does not fully characterize explanation robustness. An explanation may be substantially shifted while still remaining faithful to the model's prediction, whereas carefully optimized sparse perturbations can simultaneously alter the saliency map and reduce its faithfulness. Across Grad-CAM, HiResCAM, LayerCAM, XGrad-CAM, Raw Attention, and Attention Rollout, our results consistently support this distinction across CNN and transformer-based models. These findings suggest that saliency shift and explanation faithfulness should be considered jointly when assessing the robustness of visual explanations. More broadly, evaluating explanation robustness solely through changes in

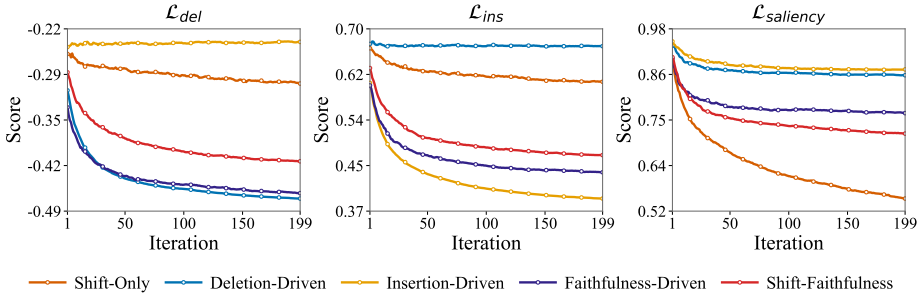


Fig. 5. Optimization trajectories of $\mathcal{L}_{\text{ins}}$, $\mathcal{L}_{\text{del}}$, and $\mathcal{L}_{\text{saliency}}$ across different attack variants. Lower values indicate more effective optimization.

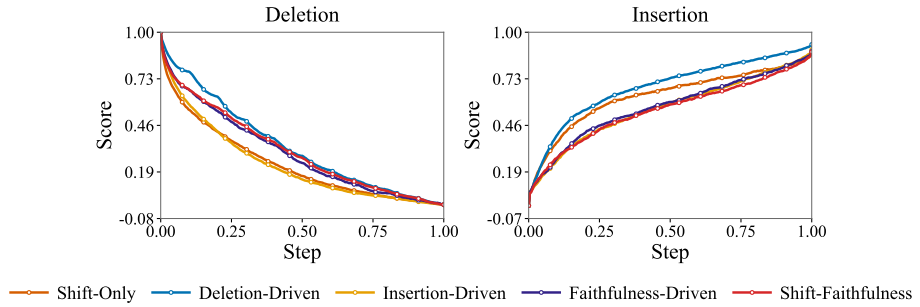


Fig. 6. Deletion and Insertion curves of the resulting explanations across different attack variants. Higher Deletion curves and lower Insertion curves indicate lower explanation faithfulness.

saliency maps may provide an incomplete view of their reliability, highlighting the importance of incorporating faithfulness into future robustness evaluations.

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